MIL-P-48618A(AR)
23 November 1983
SUPERSEDING
MIL-P-48618(AR)
22 February 1982

MILITARY SPECIFICATION

PROJECTILE, 20MM, HEI, M56A5; METAL PARTS

This specification is approved for use by the US Army Armament Research and Development Center, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the projectile metal parts for Projectile, 20MM, High Explosive Incendiary, M56A5, manufactured from bar stock by impact extrusion.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specifications For

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-651	-	Visual Inspection Standards for 20MM Ammunition and Components
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1169	-	Packaging, Packing, and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.8)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

Product and Packaging Drawings

7258816	-	Blank, Rotating Band
7258861	-	Carton, for Interplant Shipment of Projectiles, 20MM, Assemblies and Metal Parts Assemblies (200)
7259087	-	Cartridge, 20MM, for HEI Projectile Metal Parts Test
7259136	-	Shipping Container for Inert Projectile, 20MM, Assemblies and Metal Parts Assemblies (10,000)
11075857	-	Projectile, 20MM, HEI, M56A5, Metal Parts
LI 11075857	-	Index of Inspection Equipment Lists for: Projectile, 20MM, HEI, M56A5, Metal Parts
11075858	-	Body and Rotating Band Assembly, Projectile, 20MM, HEI, M56A5
11075859	-	Body, Projectile, 20MM, HEI, M56A5

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Inspection Equipment Drawings

7259353	-	Go, Flat Plug Gage, Special
7259419	-	Band Tightness Tester
7259433	-	Gage, Indicator
7259434	-	Scale
7259467	-	Comparator, Optical, Hand Type
7259489	-	Adapter, Thread Plug
7259492	-	Acceptance Check Gage
7259497	-	Go, Thread Plug Gage
7259536	-	Not Go, Flat Plug Gage, Special
7259545	-	Operating Instruction (Band Tester)
7259678	-	Gage, Indicator
7259718	-	Optical Gaging Chart
7259740	-	Gage, Snap, Special
8655592	-	Go, Ring Gage, Special
8657201	-	Optical Gaging Chart
11075081	-	Gage, Indicator
11075092	-	Gage, Ring, Plain Go, Special
11075094	-	Gage, Snap, Go, Special
11075203	-	Eddy Current Flaw Detector
11075239	-	Gage, Comparator
11075247	-	Requirements for Optical Gaging Charts
11075248	-	Block, Setting
11075270	-	Optical Gaging Chart

PUBLICATIONS

AMCR 715-505	-	Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)
ASTM B117 - Standard Method of Salt Spray (Fog) Testing
ASTM A108 - Steel Bar, Carbon, Cold Finished, Standard Quality

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.

2.1.4 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General.

a. Material. Materials shall be in accordance with the applicable drawings and specifications.

b. Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without prior approval. (See 6.5)

3.2 Components. All parts and assemblies shall comply with all requirements specified on (Dwg.) 11075857, all associated drawings, and with all requirements specified in applicable specifications.

3.3 First article sample. Unless otherwise directed by the contracting officer, a first article sample shall be required.

3.4 Banding. The rotating band blank and the rotating band seat shall be clean at the time of banding. Shot or sandblasting shall not be used to clean the rotating band seat after knurling. The rotating band blank shall fill the band seat.

3.5 Soundness. Metal discontinuity in the projectile body, as determined by eddy current inspection, shall be less than that represented by Production Check Standard (PCS) No. 1.

3.6 Protective coating.

3.6.1 Phosphate coating. The pretreatment cleaning process shall be in accordance with Dwg. 11075857. The phosphate coating shall be in accordance with the requirements of TT-C-490 for appearance and minimum weight as applicable to the type of phosphate coating used. Test specimens (panels) shall be in accordance with the requirements of TT-C-490.

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3.6.2 Paint coating. The interior and exterior coatings of the projectile metal parts shall be in accordance with applicable requirements of Dwg. 11075857. The exterior paint adhesion shall be in accordance with applicable requirements of TT-C-490.

3.6.3 Corrosion resistance. The exterior painted surfaces of the projectile metal parts shall show no evidence of corrosion (red rust) after exposure to a 5 percent solution of atomized salt spray for 24 hours.

3.7 Metal parts security. The projectile metal parts, inert charged, and assembled into cartridges in accordance with Dwg. 7259087, shall not separate or break up in the test for metal parts security.

3.8 Homogeneity. The impact extrusion process used to manufacture the projectile shall cause definitely defined metal movement as evidenced by micro and macro analysis.

3.9 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.9.1 Processing defects. The projectile metal parts shall be free of defects, such as dents, deep scratches, cuts, excessive surface roughness, metal flaws, burrs, sharp edges, and other forms of substandard workmanship.

3.9.2 Cleanliness. The projectile metal parts shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection (4.3)
b. Quality Conformance Inspection (4.4)

4.3 First article inspection.

4.3.1 Submission. After inspection by the contractor and provisional acceptance by the Government, the contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items as indicated.

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a. Inspection and nonfiring tests:

Parts description	Drawings	Quantity	Method Para
Standard test panels		3	4.5.5
Rotating bands	7258816	32	4.4.2.1
Projectile bodies	11075859	125	4.4.2.2
Projectile body assemblies (uncoated)	11075858	125	4.4.2.3
Projectile metal parts (coated prior to paint)	11075857	125	4.4.2.4
Projectile metal parts (painted)	11075857	170	4.4.2.5

b. Firing test:

Parts description	Drawings	Quantity	Method Para
Projectile, metal parts	11075857	4,000	4.5.8

c. The projectile metal parts, Dwg. 11075857, shall be identified in accordance with MIL-STD-1168.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078. Failure of the original first article sample to be in accordance with applicable drawings and the requirements of TABLE I, as specified herein, will result in sample disapproval.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48618A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Projectile, 20MM, HEI, M56A5, Metal Parts	1 OF 2	SEE BELOW
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rotating Bands - Dwg. 7258816				
	Examination of defects	32		3.2	4.4.2.1
	Body, Projectile - Dwg. 11075859				
	Examination of Defects	125		3.2	4.4.2.2
	Homogeneity radial metal flow in base section	6		3.8	4.5.9
	Body and Rotating Band Assemblies (Uncoated) - Dwg. 11075858				
	Examination of defects	125		3.2	4.4.2.3
	Soundness verification	125	0-1*	3.5	4.5.4
	Projectile, Metal Parts (coated prior to paint) - Dwg. 11075857				
	Examination of defects	125		3.2	4.4.2.4
	Standard test panels (specimens)	3	0-1*	3.6.1	4.5.5

NOTES:

*Accept-Reject Criteria.

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48618A(AR)

PARAGRAPH	TITLE			SHEET 2 OF 2	DRAWING NUMBER
	Projectile, 20MM, HEI, M56A5, Metal Parts				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile, Metal Parts (painted) - Dwg. 11075857				
	Examination of defects	125		3.2	4.4.2.5
	Band tightness	125	0-1*	3.4	4.5.1/7259419
	Paint adhesion	13	0-1*	3.6.2	4.5.6
	Corrosion resistance	32	0-1*	3.6.3	4.5.7
	Firing Test				
	Metal parts security test	4000	0-1*	3.7	4.5.8

NOTES:

*Accept-Reject Criteria.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. To determine product acceptability, major and minor defects as listed herein may be considered collectively on a class basis or defects may be considered individually. When a class basis is utilized the Acceptable Quality Level (AQL) for major defects shall be 1.50 percent and the AQL for minor defects shall be 2.50 percent. However, when major defects are considered collectively on a class basis, the AQL for any individual defect shall be limited to .40 percent. In addition, when three or less defects appear in a classification, sampling plans applicable to a class shall not be used for major defects. (See 6.7)

c. MIL-STD-651 shall apply in defining and evaluating visual defects.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48618A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Rotating Band Blank	1 OF 1	7258816
			NEXT HIGHER ASSEMBLY: 11075857

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101.	Foreign matter		.25%	3.9.2	Visual
102.	Defective metal, including lamination		.25%	3.2	Visual
103.	Incomplete or deformed		.25%	3.2	Visual 1/
MINOR					
201.	Width, max.		.40%		11075094
202.	Evidence of poor workmanship		.65%	3.9	Visual

NOTES: 1/ A rotating band blank which is deformed or is judged to lack sufficient metal in any part of its width or thickness to provide for complete band seat fill and minimum rotating band profile dimensions, shall be classed defective.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48618A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Body, Projectile	1 OF 1	11075859
			NEXT HIGHER ASSEMBLY: 11075857

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101.	Length, mouth to rear edge of rotating band seat		.25%	3.2	7259740
102.	Knurling missing or incomplete		.25%	3.2	Visual
103.	Depth, knurling, min.		.25%	3.2	Visual/7259678 1/
104.	Foreign matter in rotating band seat		.25%	3.4	Visual
MINOR					
201.	Width, rotating band seat		.40%	3.2	7259353/7259536
202.	Diameter, rotating band seat knurling		.40%	3.2	Gage 2/
203.	Width, knurling, min.		.40%	3.2	Visual/7259467 1/
204.	Evidence of poor workmanship		.65%	3.9	Visual

NOTES:

1/ Measurement of knurling for compliance with the minimum depth and width requirements shall apply where visual inspection indicates that the depth or width dimension is borderline.

2/ Gage, Snap - (Anvils, .170 min. width)

(Go .680) .6796 - .0004

(Not Go .662) .6620 + .0004

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48618A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Body and Rotating Band Assembly, Uncoated	1 OF 3	11075858/11075859
			NEXT HIGHER ASSEMBLY: 11075857

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1.	Soundness-ferrous metal discontinuity test		100%	3.5	4.5.2/11075203
MAJOR					
101.	Diameter, bourrelet, min.		.25%	3.2	Gage 1/
102.	Diameter, body, rear of rotating band, min.		.25%	3.2	Gage 2/
103.	Run out, body, rear of rotating band, with datum surface		.25%	3.2	7259433/7259492/ 7259489
104.	Diameter, rotating band		.25%	3.2	Gage 3/
105.	Length, mouth to rear edge of rotating band, max.		.25%	3.2	11075239/11075248
106.	Run out, rotating band with datum surface		.25%	3.2	7259433/7259492/ 7259489

NOTES:

1/ Snap Gage - (Not Go .782) .7820 + .0001
2/ Snap Gage - (Not Go .765) .7650 + .0003
3/ Ring Gage - (Go .828) .82780 - .00012
Snap Gage - (Not Go .824) .8240 + .0002

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Body and Rotating Band Assembly, Uncoated	2 OF 3	11075858/11075859
			NEXT HIGHER ASSEMBLY: 11075857

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
107.	Length, to internal basic diameter		.25%	3.2	Gage
108.	Profile, crimp groove		.25%	3.2	Comparator/20x/8657201
109.	Radius, rear edge of crimp groove, max.		.25%	3.2	Comparator/50x
110.	Location, crimp groove		.25%	3.2	Comparator/20x/8657201
111.	Wall thickness variation, max.		.25%	3.2	Gage
112.	Thickness, base, min.		.25%	3.2	11075081 4/
113.	Thickness, wall, min.		.25%	3.2	Gage
114.	Total length, max.		.25%	3.2	Gage
115.	Void, junction rotating band and body		.25%	3.4	Visual
116.	Threads not complete		.25%	3.2	Visual
117.	Base face machined		.25%	3.2	Visual
118.	Radius, base of projectile		.25%	3.2	Visual/Comparator 5/

NOTES:

4/ Base thickness shall be measured at the approximate longitudinal axis from opposite surface flats. A body having base thickness less than the specified minimum shall be classed defective.

5/ Measurement of mouth-ogive break and the base radius shall apply where visual inspection indicates that the maximum dimension is borderline.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48618A(AR)

PARAGRAPH	TITLE	SHEET 3 OF 3	DRAWING NUMBER
4.4.2.3	Body and Rotating Band Assembly, Uncoated		11075858/11075859
			NEXT HIGHER ASSEMBLY 11075857

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201.	Depth, thread counterbore (if present), max.		.40%	3.2	Visual/Gage 6/
202.	Diameter, thread counterbore or chamfer (if present)		.40%	3.2	Visual/Gage 6/
203.	Outside diameter, mouth end		.40%	3.2	11075270/Comparator/ 10x
204.	Total length, min.		.40%	3.2	Gage
205.	Length, mouth to rear edge of rotating band, min.		.40%	3.2	11075239/11075248
206.	Finish improper		.40%	3.2	Visual
207.	Break, mouth, at ogive, max.		.40%	3.2	Visual/11075270/ Comparator/10x
208.	Depth and width of undercut front and rear of rotating band		.40%	3.2	7259718/Comparator/ 31.25x
209.	Mutilated		.40%	3.2	Visual
210.	Corrosion		.40%	3.2	Visual
211.	Width, rotating band		.40%	3.2	Comparator/20x/8657201
212.	Evidence of poor workmanship		.65%	3.9	Visual

NOTES: 6/ Measurement of the diameter and depth, max., of the thread counterbore or chamfer shall apply where visual inspection indicates that the diameter or depth, max., dimension is borderline.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48618A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11075857
4.4.2.4	Projectile, Metal Parts (Coated, Prior to Painting)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101.	Pitch diameter, thread		.25%	3.2	7259497 1/
102.	Run out, thread with datum surface		.25%	3.2	7259433/7259492/ 7259489
MINOR					
201.	Minor diameter, thread		.40%	3.2	Gage 2/
202.	Length, complete thread, min.		.40%	3.2	Gage 1/
203.	Evidence of poor workmanship		.65%	3.9	Visual

NOTES: 1/ Thread Plug Gage - Go .5422 and Thread length, min.
Thread Plug Gage, expanding type (cone and vee) with stop*, Not Go .5469.
Thread Set Check - Thread Ring Gage (.5469 - .0001)

2/ Plain Plug Gage, expanding type with stop*
.5287 + .0057 (max. anvil width .080)
Thread Set Check, Plain Ring Gage (.53155 ± .00005)

* Stop shall be suitable to check thread throughout min. thread depth specified on component drawing.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48618A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Projectile, Metal Parts (Painted)		11075857
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101.	Diameter, bourrelet, max.		.25%	3.2	8655592
102.	Diameter, body, rear of rotating band, max.		.25%	3.2	11075092
103.	Weight		.25%	3.2	Balance/7259434
104.	Foreign matter in cavity or threads		.25%	3.2	Visual 1/
MINOR					
201.	Interior or exterior paint inadequate		.40%	3.2	Visual
202.	Exterior paint not dry		.40%	3.2	Manual
203.	Paint on rotating band		.40%	3.2	Visual
204.	Paint on threads		.40%	3.2	Visual
205.	Marking incorrect, incomplete, illegible or missing		.40%	3.2	Visual
206.	Rotating band damaged		.40%	3.2	Visual
207.	Foreign matter on exterior		.40%	3.2	Visual
208.	Evidence of poor workmanship		.65%	3.9	Visual

NOTES:

1/ Particular attention shall be given to examination of the cavity for steel chips and shavings.

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4.4.3 Tests. The tests listed in Table II shall be conducted on each lot in compliance with the methods of inspection prescribed in 4.5. The samples shall be selected in such a manner as to be representative of the entire lot. The sample size and acceptance criteria for each test shall be as specified.

TABLE II

Test		Sample Size	Item	Rqmt Para
Band tightness/7259419	1/	315	Projectile, metal parts (See 4.5.1)	3.4
Soundness verification	2/	315	Projectile, metal parts, unpainted (See 4.5.4)	3.5
Automatic inspection effectiveness	3/		PCS No. 1 (See 4.5.3)	
Phosphate coating (Appearance & weight)	4/	3	Projectile, metal parts (before painting/or standard panels) (See 4.5.5)	3.6.1
Paint adhesion	5/	13	Projectile, metal parts, TT-C-490 (See 4.5.6)	3.6.2
Corrosion resistance	6/	32	Projectile, metal parts (See 4.5.7)	3.6.3
Metal parts security	7/	250	Projectile, metal parts (See 4.5.8)	3.7

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1/ If two or more units of the sample have an average radial void between the seated band and the band seat, greater than 0.004 inch, the lot shall be rejected. If one unit of the sample has an average radial void, greater than 0.004 inch, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units have an average radial void greater than 0.004 inch, the lot shall be rejected.

2/ Rejection of one or more units of the sample by the flaw detector in the test for body metal soundness shall be cause for rejection of the lot.

3/ Failure of the flaw detector to reject PCS No. 1 shall be cause for rejection of all projectile metal parts inspected for "soundness" since the preceding test and shall require requalification of the equipment for automatic inspection. (See 4.5.3).

4/ Failure of the phosphate coating on the sample projectiles or standard panels to comply with the applicable requirements shall cause rejection of all production projectile metal parts phosphated since the preceeding test. (See 3.6.1).

5/ If two or more units of the sample fail to comply with the requirement for paint adhesion, the lot shall be rejected. If one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the requirement for paint adhesion, the lot shall be rejected.

6/ If two or more units of the sample fail to comply with the corrosion resistance requirement, the lot shall be rejected. If one unit fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the corrosion resistance requirement, the lot shall be rejected.

7/ The lot will be rejected, if any one of the following types of metal parts separation or breakup occurs in the metal parts security test, as witnessed by hole(s) in the fragmentation screen(s), or recovery of the part(s) or fragment(s).

a. Body breakup.

b. Complete rotating band separation. This is defined as detachment of all of the band having band seat knurling imprint.

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c. Partial rotating band separation. This is defined as detachment of part of the band having band seat knurling imprint.

d. The following acceptance criteria shall apply to thrown band fragments having no evidence of band seat knurling imprint:

(1) The lot will be accepted if two or less projectile metal parts throw band fragments having length not greater than 0.500 inch.

(2) The lot will be rejected if three or more projectile metal parts throw band fragments having length not greater than 0.500 inch.

(3) The lot will be rejected if any projectile metal parts throws a band fragment having length greater than 0.500 inch (indicative of deep-seated band lamination).

(4) The lot will not be penalized for band particles thrown as a result of normal band fringing or slivering.

4.4.4 Packaging, packing and marking inspection. Inspection for packaging, packing and marking shall comply with applicable requirements of MIL-STD-1169.

4.4.5 Inspection equipment. The Inspection Equipment required to perform the examinations and tests prescribed herein is described in the Index of Inspection Equipment List - LI 11075857 and the paragraph reference/inspection method column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Band tightness. The method of test shall comply with all requirements of the operating instructions for the band tightness tester Dwg. 7259545. The inspection sample shall be cycled through the tester to determine compliance with the rotating band tightness requirement. The total measurement of radial void registered on the tester indicator at final ram pressure, shall be divided by four (number of indentors) to determine the average radial void.

4.5.2 Soundness. The method of inspection shall comply with all requirements of Eddy Current Flaw Detector, Dwg. 11075203, except for those inspection requirements which are reserved for Government application, as specified herein.

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4.5.2.1 Application of defect standards. Master defect standards, referenced in the Operating Instructions, shall apply to resolve inspection discrepancies occurring between the results of defect standards supplied for contractor "soundness" inspection and those supplied for Government "soundness" verification. (See 6.6).

4.5.3 Automatic eddy current inspection effectiveness. Equipment checkout shall be performed as stated in the operating instructions at the start of each shift and after any adjustment or shutdown. Upon completion of checkout, cycle the PCS No. 1, front slot and rear slot standard through the inspection system during automatic operation. All three standards must be rejected to assure correct adjustment for reject level and scan limits. During remainder of shift the PCS No. 1 shall be cycled through the equipment at intervals of 20 $\pm$ 5 minutes during automatic operation.

4.5.3.1 Equipment verification. The Government inspector will maintain independent of the contractor, a set of standards and may cycle these standards through the equipment at any time. Failure of equipment to reject standards will be evidence that the equipment is not properly adjusted and the contractor will be so notified.

4.5.4 Soundness verification. After equipment checkout and verification as specified in 4.5.3 and 4.5.3.1 the entire lot sample will be inspected with the equipment operating automatically. Any rejected unit of the sample will be automatically cycled through the equipment with the CAL. button pressed (to lower the sensitivity one step below that used for production inspection). Any unit which is rejected at the lower sensitivity shall be classed as a defective.

4.5.5 Phosphate coating. The method of test shall be as prescribed in TT-C-490 for the type of coating used (see Dwg. 11075857). The prescribed sample projectiles or standard panels shall be processed at the start of daily production and at 4-hour intervals, maximum, thereafter. Appearance and weight of the coating shall be recorded.

4.5.6 Paint adhesion. The method of test shall be as prescribed in TT-C-490. The test shall be made on the exterior painted surface of each projectile metal part of the sample forward of the rotating band. Visible exposure of underlying phosphate coating or ferrous metal on any unit of the sample, as a result of the test, shall be recorded as a defect.

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4.5.7 Corrosion resistance. The method of test shall be as prescribed in American Society for Testing and Materials (ASTM), ASTM B117. The paint shall not be disturbed, however, the rotating band and unpainted surfaces of the sample projectile metal part may be masked out by a wax coating or other suitable material. The salt spray solution and exposure time shall be as specified in 3.6.3.

4.5.8 Metal parts security. The method of test will be as prescribed in AMCR 715-505, Volume 8, as applicable. The projectile metal parts will be charged and assembled as specified on Dwg. 7259087, Cartridge, 20MM, for projectile metal parts test. The test cartridges, at ambient temperature, will be fired in bursts of 50 shots in an M39 gun. The gun barrel will be at ambient temperature at the beginning of the test and cooled to ambient temperature after each burst. The cyclic rate of fire will be recorded for informational purposes.

4.5.9 Homogeneity. The contractor shall demonstrate by micro and macro analysis of samples from the First Article lot that sufficient radial metal flow in the base section has occurred during the impact extrusion of the projectile. The contractor shall submit his report on the homogeneity of the first article sample to Commander: ARDC, ATTN: DRSMC-QAF-S(D) and DRSMC-SCA-CE(D), Dover, NJ 07801 for acceptance. (See 6.6.1)

5. PACKAGING

5.1 Preservation and Packaging.

5.1.1 The projectile metal parts shall be preserved and packaged in accordance with Dwg. 7258861 or Dwg. 7259136, as directed by the procuring activity.

5.2 Packing.

5.2.1 The projectile metal parts, preserved and packaged as specified in 5.1. Each lot shall be packed for shipment in accordance with Dwg. 7258861 or Dwg. 7259136, as directed by the procuring activity.

5.3 Marking.

5.3.1 Marking or labeling of packed cartons shall comply with applicable requirements of MIL-STD-1169.

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6. NOTES

6.1 Intended use. The component covered by this specification is intended for use with Cartridges, 20MM, HEI, M56A3 and M56A4.

6.2. Ordering data. See MIL-A-48078.

6.3. Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ATTN: DRSMC-QAF-I(D), US Army Armament Research and Development Center (ARDC), Dover, NJ, 07801.

6.4 First article sample. Where a first article sample is required by the contract, the contracting officer will notify the test facility when to proceed with the metal parts security test. Additional assemblies may be required by the test facility. (See 4.3)

6.5 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item.

6.6 Defect standards. The initial supply of defect standards and instructions for contractor and Government application to the eddy current method of inspection will be furnished. Replacement standards should be requisitioned through normal channels by the Government agency having contract administration responsibility.

6.6.1 Homogeneity standards. (Samples) See Figures 1 and 2.

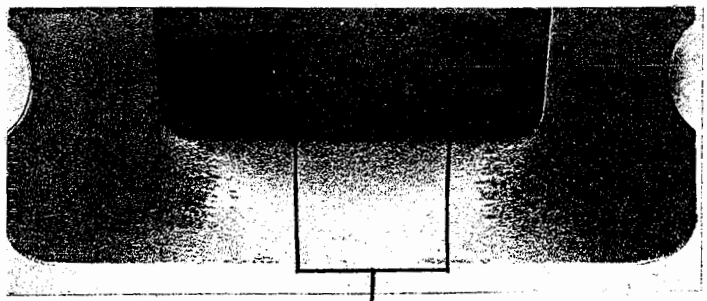
6.7 Intermediate point inspection. The classification of defects identifies the defect characteristics (among other things) for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which would be detrimental to the attainment of optimum product quality of the end item.

6.8 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or U.S. Army Armament Research and Development Command (ARRADCOM) drawings. Technical data originally prepared by these activities are now under the cognizance of ARDC.

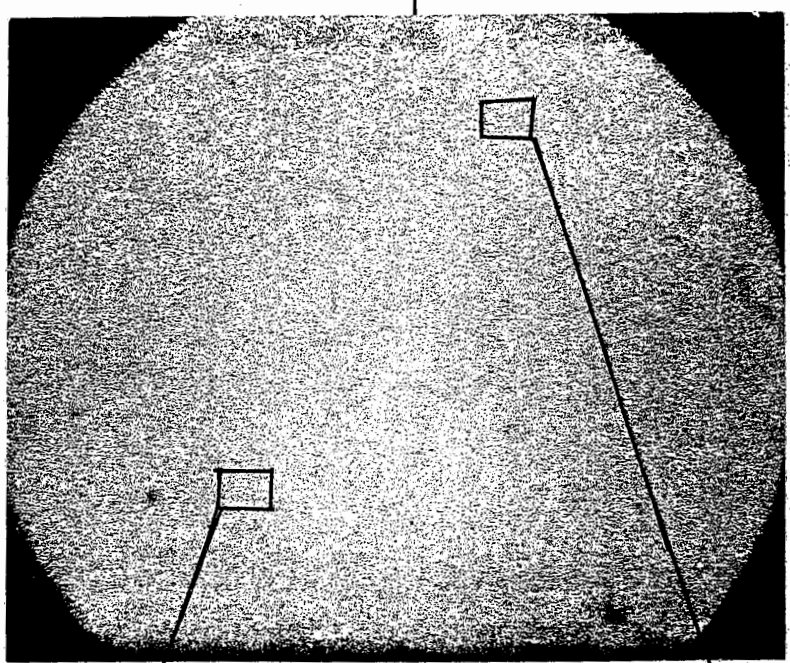
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6.9 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

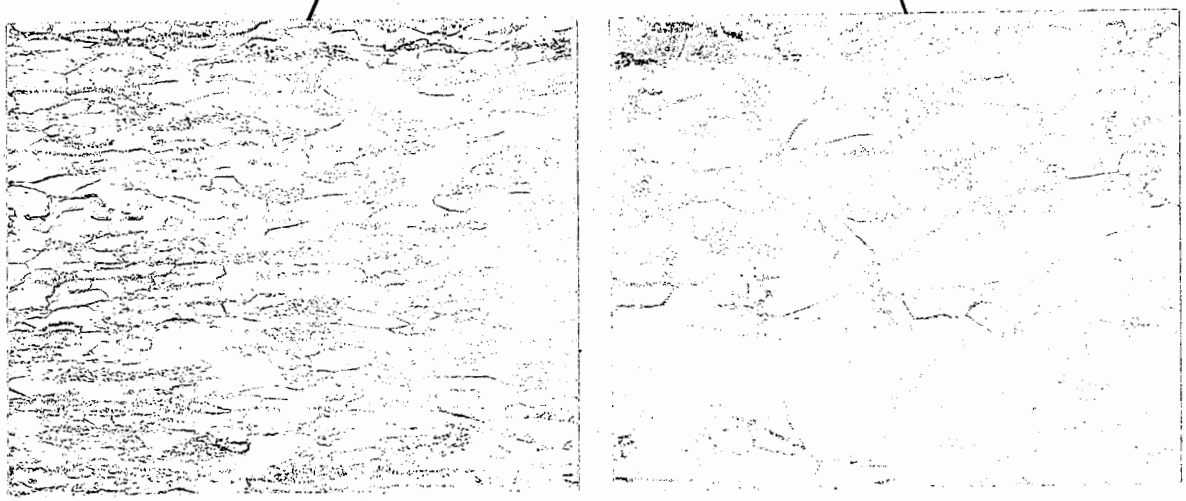
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5X



25X

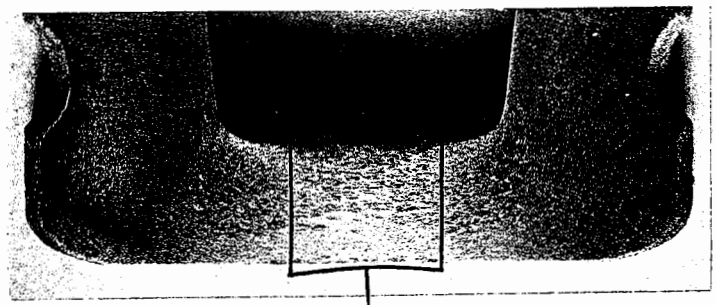


400X

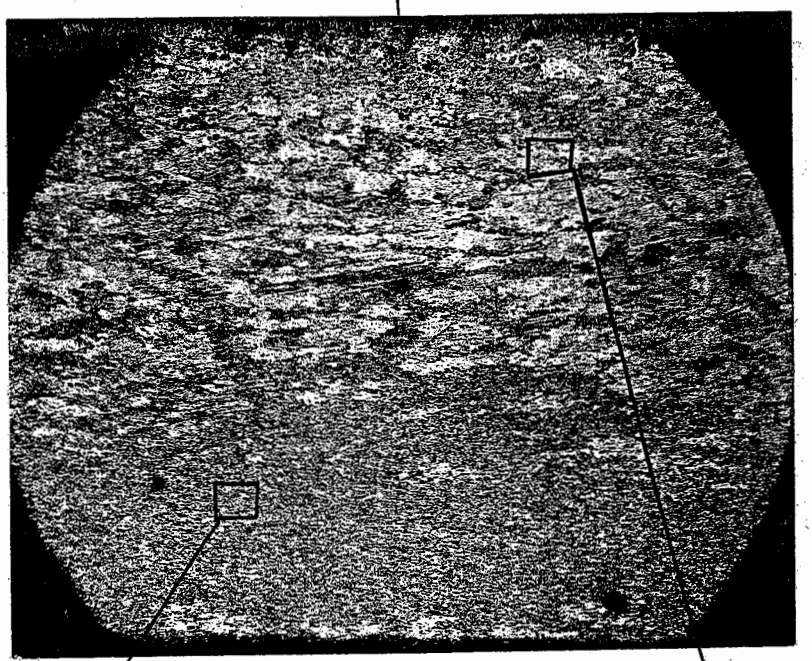
Structure in the base of 20mm M56A5 projectile #1 manufactured by impact extrusion. Radial flow is observed at 5X at the base wall radius where large grains are present. The finer grains toward the outside surface of the base must be observed at high magnification (400X) to show the radial elongation of the grains.

FIGURE 1

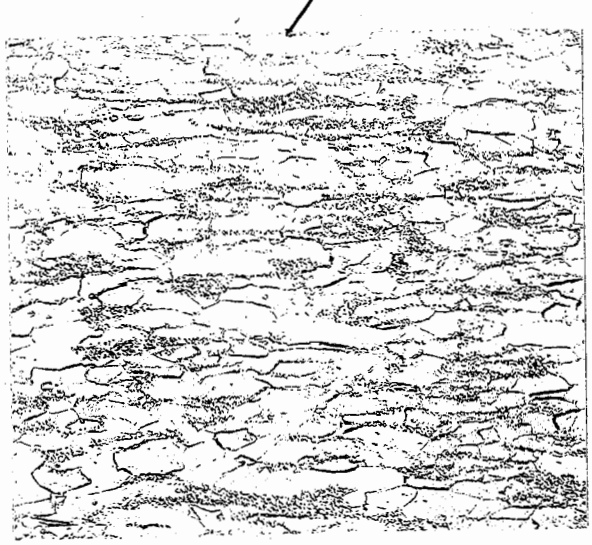
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5X



25X





400 X

Structure in the base of 20mm M56A5 projectile #3 manufactured by impact extrusion. Radial flow is more apparent at low magnification than in projectile #1 because of the greater abundance of large grains. In other respects projectile #3 is similar to #1.

FIGURE 2

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Preparing Activity:
Army - AR

(Project 1305-A950)

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